

Assignment 3

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Code Implementation:

-The primary algorithm prepared for by this code is Stereo Correspondence It attempts to find the same point in two different images

-The process works by shifting a small window (block) from the left image across the corresponding row in the right image to find the best match based on pixel intensity.

-The fundamental calculation performed at the pixel level is Disparity which represents the horizontal displacement of an object between the two views:

$$d = x_{\text{left}} - x_{\text{right}}$$

Where x is the horizontal coordinate of a feature in the respective image planes.

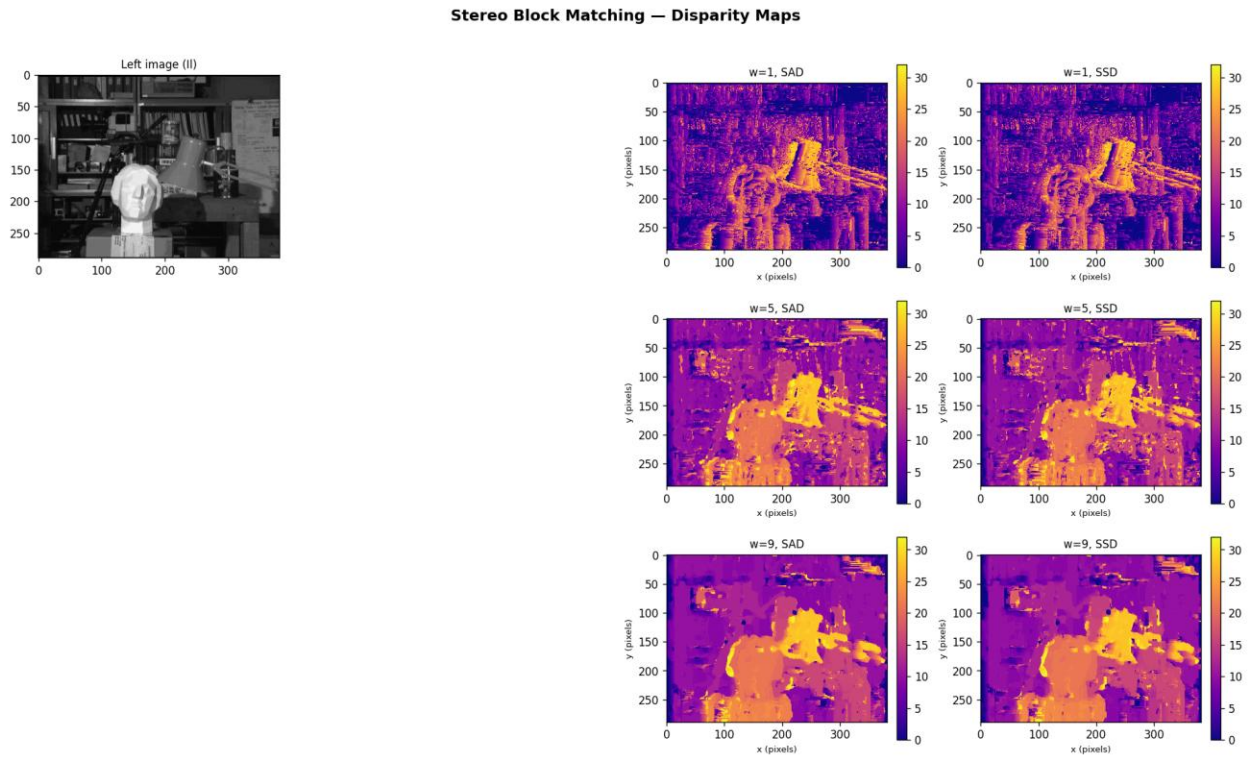
-Once the disparity is calculated, the system can derive physical distance Z from the camera This is based on the inverse relationship between shift and distance:

$$Z = (f * B) / d$$

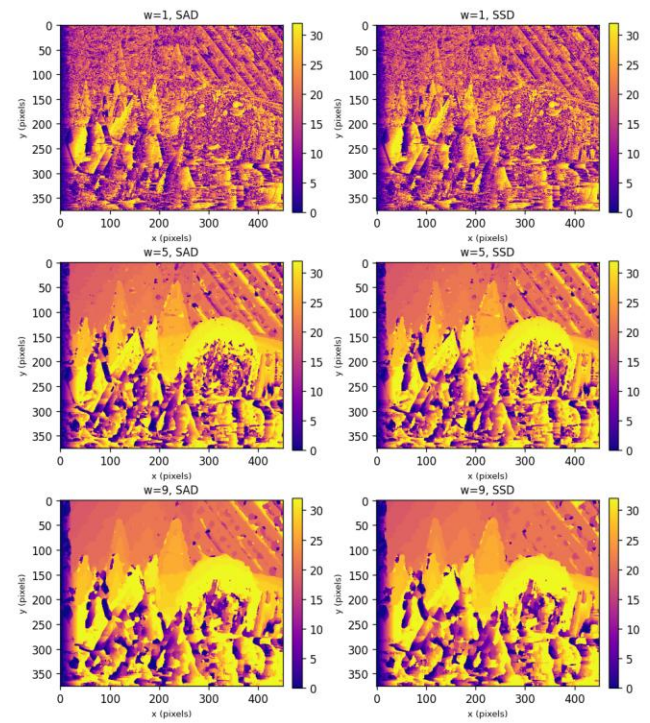
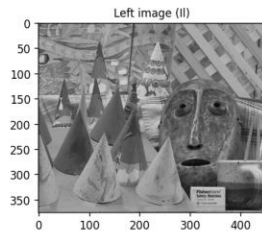
- f (Focal Length): The distance from the camera lens to the sensor.

- B (Baseline): The physical distance between the two camera centers.
- d (Disparity): The value derived from the matching algorithm.

Test Runs:



Stereo Block Matching — Disparity Maps



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